Executive Summary - Coalition Warehouse Database Project:
http://classwork.engr.oregonstate.edu:30905/

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Step 1 – Project Proposal, Outline, ERD:

Feedback – Redundant attributes within Customers and Vendors table. Naming convention ambiguity for productCost vs vendorCost. Inconsistent naming conventions in regards to PascalCase vs camelCase. Insufficient outline distinction between entities and intersection tables.

Changes – Removed vendorNumber and customerNumber attributes from their respective tables. Renamed productCost to listPrice within the Products table. Aligned all tables to use PascalCase and all attributes to use camelCase. Better organized the outline to distinguish between entities and transaction/intersection tables. Added a purpose section to clarify the use/role of each entity. Lastly, renamed productCost to listPrice in Products table and itemPrice to unitPrice in PurchaseOrderItems table.

Step 2 – Normalized Schema + DDL with Sample Data

Feedback – Missing ON DELETE CASCADE on Inventory and VendorProducts. Inconsistency between outline and DDL with definition of purchaseDate attribute. Inconsistency of plural and singular table names.

Changes – Added ON DELETE CASCADE to Inventory and VendorProducts tables. Aligned all time stamps to be CURRENT_TIMESTAMP. Fixed naming inconsistency of table names to be all plural.

Step 3 – Design UI Interface + DML SQL:

Feedback – Non-functional UPDATE/CREATE/DELETE buttons. Low contrast of navigational bar.

Changes – Button functionality added at a later step. Nav bar made larger and increased its contrast to align with the rest of the page. Added direct links to all other pages on the homepage.

Step 4 – Stored Procedures and SELECTs for all entities:

Feedback – Routing bug allowing for old JSON data to be displayed after CRUD operations. Discrepancy between DB and Sales page for Total Cost column. SP naming convention inconsistencies.

Changes – Fixed routing issue displaying JSON. Fixed DB and Sales page discrepancy. Added window.confirm to avoid accidental deletions. Renamed Sps to start with sp_ and triggers with tr_.

Step 5 – CUD operations for all tables:

Feedback – Issue with deleting the node_modules folder caused the website to break and no DB operations to work.

Changes – Ran NPM install and it fixed the issue. Will make sure to re-run it and test the website after creating the final zip file with our code to avoid similar issues for final submission.

Reflection on AI Usage:

Throughout this project, we used Gemini Code Assist (VS Code extension) for debugging and feature scaffolding. It was most helpful for initial deployment of the project, identifying issues in our Express routes and suggesting patterns for expandable table components. However, AI-generated code often required manual review and adaptation to fit our existing code. Overall I think it was quite helpful and allowed us to accomplish more without over relying on it.

Project Outline

Overview:
The owner of Coalition Warehouse LLC has been in business for over a decade. In that time, he has seen record growth and expansion of his business. He has been running this business solely with the use of Excel spreadsheets to keep track of vendor orders, warehouse inventory, products, customers, and customer orders. Due to this expansion to multiple locations, he has been plagued with inventory stocking issues, such as too much inventory or too little inventory.

Coalition Warehouse LLC sees ~40,000 transactions yearly, encompassing 500 products with 50 international vendors and 10,000 customers. A database-driven website will allow for tracking records of inventory of products within his warehouses. Additionally, tracking records of purchases from vendors and sales to customers. Lastly, allow Coalition Warehouse to make informed decisions on ordering trends for the future.

Database Outline:

Entities:

- Products:
- Purpose: Records the product information stocked by Coalition Warehouse LLC
- productID: int(11), auto_increment, unique, not NULL, PK
- productName: varchar(255), not NULL
- listCost: decimal(10,2), not NULL
- Relationship:
- 1:M - VendorProducts (M:N with Vendors)
- 1:M - Inventory
- 1:M - SalesOrderItems (M:N with SalesOrders)
- 1:M - PurchaseOrderItems (M:N with Purchase_Orders)

- Vendors:

- Purpose: Records the details of vendors Coalition Warehouse LLC gets products from.
- vendorID: int(11), auto_increment, unique, not NULL, PK
- vendorName: varchar(255), not NULL
- vendorAddr: varchar(255)
- vendorEmail: varchar(255)
- Relationship:
- 1:M - VendorProducts (M:N with Products)
- 1:M - PurchaseOrders
- Warehouses:
- Purpose: Records the warehouses owned by Coalition Warehouse LLC
- warehouseID: int(11), auto_increment, unique, not NULL, PK
- warehouseName: varchar(255), not NULL
- warehouseAddr: varchar(255)
- Relationship:
- 1:M - Inventory
- 1:M - PurchaseOrders
- 1:M - SalesOrders

- Customers:
- Purpose: Records the personal and contact information of customers of Coalition Warehouse LLC
- customerID: int(11), auto_increment, unique, not NULL, PK
- customerLN: varchar(255), not NULL
- customerFN: varchar(255), not NULL
- customerAddr: varchar(255), not NULL
- customerEmail: varchar(255)
- Relationship:
- 1:M Sales_Orders

Transaction Tables:

- SalesOrders:
- Purpose: Records individual sales by Coalition Warehouse LLC to customers
- saleOrderID: int(11), auto_increment, unique, not NULL, PK
- customerID: int(11), NOT NULL, FK
- warehouseID: int(11), NULL, FK
- saleDate: DATETIME, default CURRENT_TIMESTAMP, not NULL
- Relationship:
- M:1 - Customers
- M:1 - Warehouses
- 1:M - SalesOrderItems (M:N with Products)
- PurchaseOrders:
- Purpose: Records purchases from vendors by Coalition Warehouse LLC to restock inventory
- purchaseOrderID: int(11), auto_increment, unique, not NULL, PK
- vendorID: int(11), NULL, FK
- warehouseID: int(11), NULL, FK
- purchaseDate: DATETIME, default CURRENT_TIMESTAMP, not NULL
- Relationship:
- M:1 - Vendors
- M:1 - Warehouses
- 1:M - PurchaseOrderItems (M:N with Products)

Intersection Tables:

- Inventory:
- Purpose: Records the quantity of a specific product stored at a specific warehouse.
- inventoryID: int(11), auto_increment, unique, not NULL, PK
- productID int(11), not NULL, FK
- warehouseID int(11), not NULL, FK
- The combination of productID and warehouseID should be unique
- quantity: int(11), not NULL
- Relationship:
- M:1 - Products
- M:1 – Warehouses

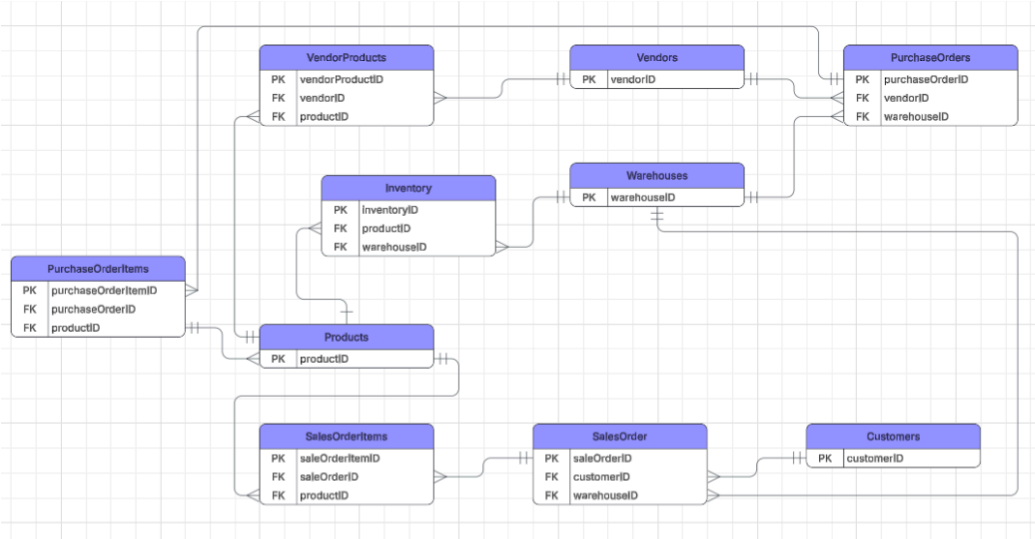
- VendorProducts:

- Purpose: Records which products are supplied by which vendors and the associated wholesale cost catalog.
- vendorProductID: int(11), auto_increment, unique, not NULL, PK
- vendorID: int(11), not NULL, FK
- productID: int(11), not NULL, FK
- costFromVendor: decimal(10,2), not NULL
- Relationship:
- M:1 - Vendors
- M:1 - Products

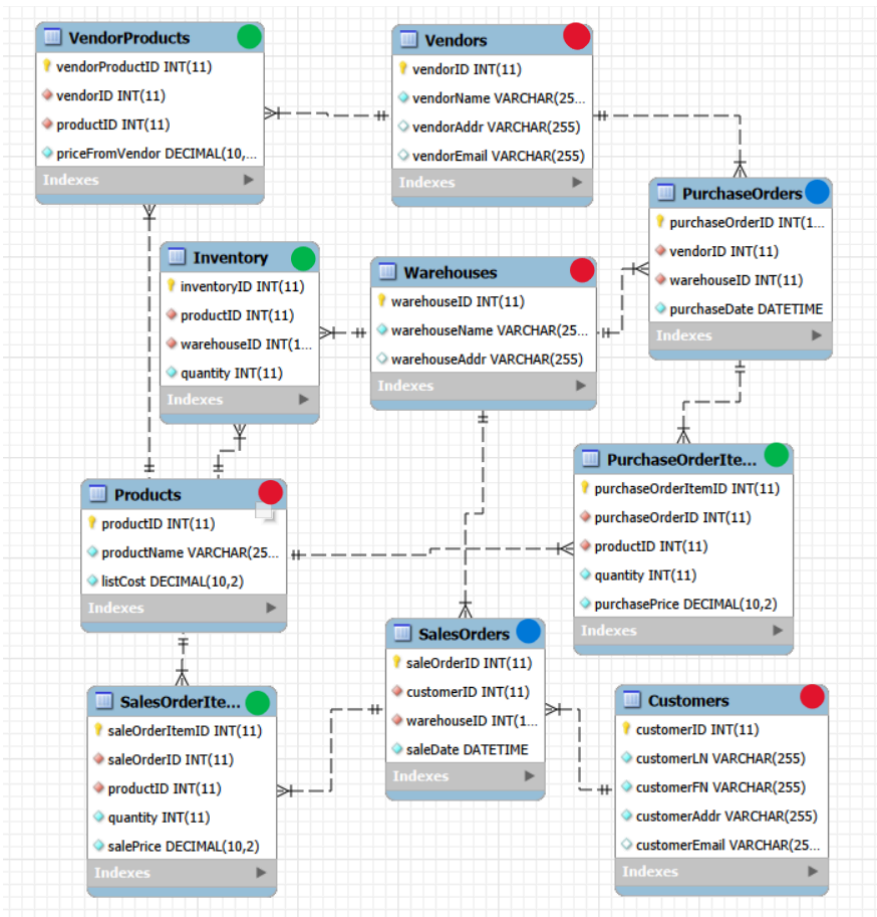
- PurchaseOrderItems:
- Purpose: Records the specific items and quantities included in a restocking order.
- purchaseOrderItemID:int(11), auto_increment, unique, not NULL, PK
- purchaseOrderID: int(11), not NULL, FK
- productID: int(11), NULL, FK
- quantity: int(11), not NULL
- purchasePrice: decimal(10,2), not NULL
- Relationship:
- M:1 - PurchaseOrders
- M:1 - Products

- SalesOrderItems:
- Purpose: Records the specific items and quantities included in a specific sales order.
- saleOrderItemID: int(11), auto_increment, unique, not NULL, PK
- saleOrderID: int(11), not NULL, FK
- productID: int(11), NULL, FK
- quantity: int(11), not NULL
- salePrice: decimal(10,2), not NULL
- Relationship:
- M:1 - SalesOrders
- M:1 – Products

Entity-Relationship Diagram:



Schema Diagram:



Sample Data:

PRODUCTS

productID	productName	listCost
1	Premium Keyboard 591	299.89
2	Plastic Drawer 653	297.88
3	Steel Desk 473	441.74
4	Ergonomic Mouse 654	220.62

VENDORS

vendorID	vendorName	vendorAddr	vendorEmail
1	Martin GmbH	602 Road 93, Dublin, Ireland	sales@martingmbh.com
2	King Group	429 Road 86, Munich, Germany	sales@kinggroup.com
3	Nelson Group	344 Street 70, Singapore, Singapore	sales@nelsongroup.com
4	White Ltd	542 Strasse 81, Seoul, South Korea	sales@whiteltd.com
5	Carter Group	829 Ave 19, Amsterdam, Netherlands	sales@cartergroup.com

VENDORPRODUCTS

vendorProductID	vendorID	productID	costFromVendor
1	4	1	96.75
2	3	2	60.62
3	2	3	84.09
4	1	4	9.90

CUSTOMERS

customerID	customerLN	customerFN	customerAddr	customerEmail
1	Hernandez	Anna	213 Park Rd, Los Angeles, CA	anna.hernandez1@example.com
2	White	Brenda	493 Elm St, Houston, TX	brenda.white2@example.com
3	Campbell	Amanda	229 Pine St, Philadelphia, PA	amanda.campbell3@example.com
4	Martin	Betty	389 Lakeview Dr, San Jose, CA	betty.martin4@example.com

WAREHOUSES

warehouseID	warehouseName	warehouseAddr
1	North Distribution	123 Frozen Lane, Anchorage, AK
2	South Fulfillment	456 Sunny Blvd, Miami, FL
3	East Coast Hub	789 Metro Way, New York, NY
4	West Coast Depot	101 Tech Drive, San Francisco, CA

INVENTORY

	inventoryID	productID	warehouseID	quantity
Edit Copy Delete	1	1	2	414
Edit Copy Delete	2	2	3	334
Edit Copy Delete	3	3	1	189
Edit Copy Delete	4	4	4	445

SALESORDERS

	saleOrderID	customerID	warehouseID	saleDate
Edit Copy Delete	1	1	4	2024-12-05
Edit Copy Delete	2	3	3	2024-05-20
Edit Copy Delete	3	2	1	2024-12-12
Edit Copy Delete	4	4	2	2024-02-08

SALESORDERITEMS

	saleOrderItemID	saleOrderID	productID	quantity	salePrice
Edit Copy Delete	1	1	4	4	250.00
Edit Copy Delete	2	4	3	5	450.00
Edit Copy Delete	3	3	2	5	310.00
Edit Copy Delete	4	2	1	2	320.00

PURCHASEORDERS

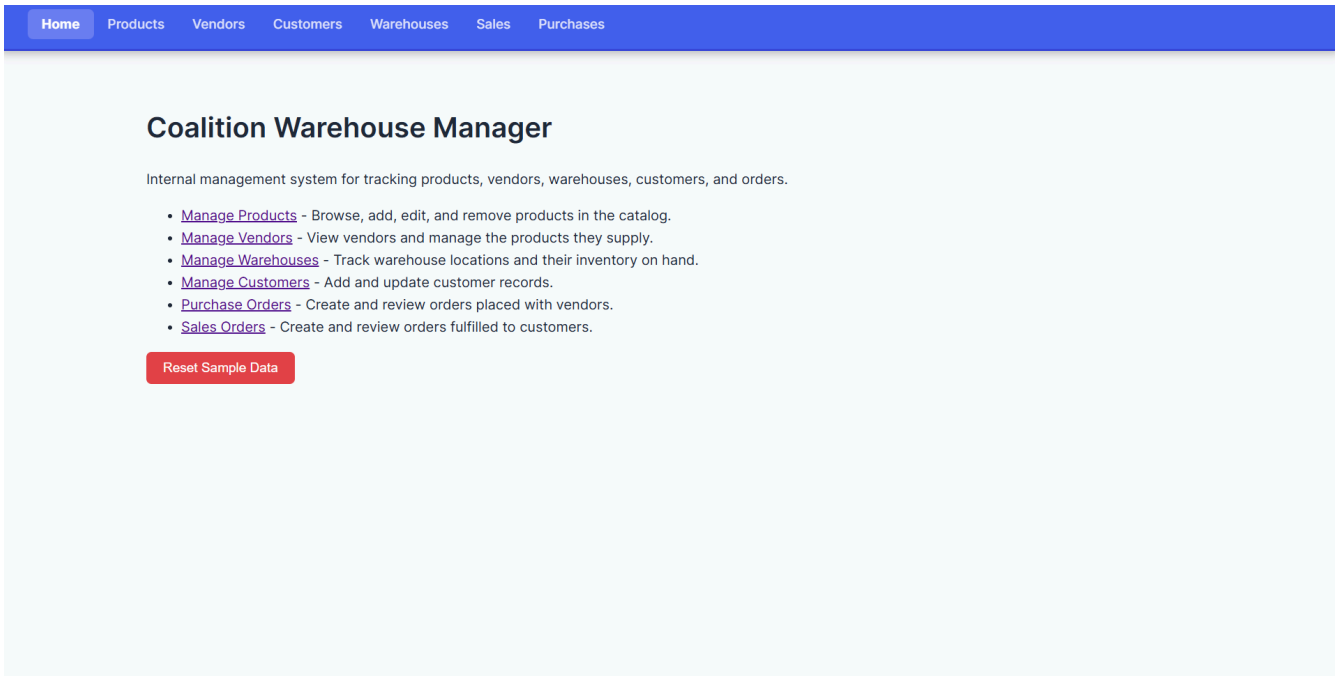
	purchaseOrderID	vendorID	warehouseID	purchaseDate
Edit Copy Delete	1	4	1	2023-11-19
Edit Copy Delete	2	1	3	2024-09-12
Edit Copy Delete	3	3	4	2024-05-12
Edit Copy Delete	4	2	1	2023-07-31

PURCHASEORDERITEMS

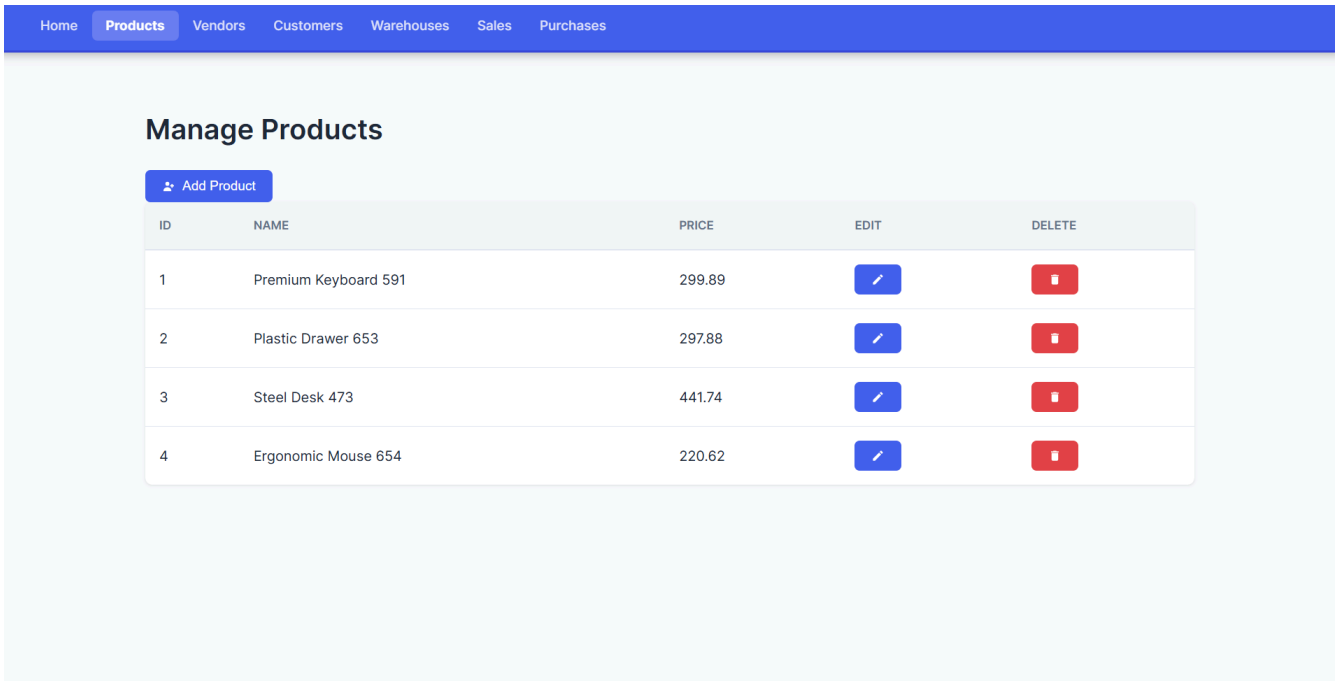
	purchaseOrderItemID	purchaseOrderID	productID	quantity	purchasePrice
Edit Copy Delete	1	1	4	98	9.90
Edit Copy Delete	2	2	3	71	84.09
Edit Copy Delete	3	3	2	92	60.62
Edit Copy Delete	4	4	1	45	96.75

UI PAGES

Homepage - RESET DB (restores DB using a SP when the red button is clicked):



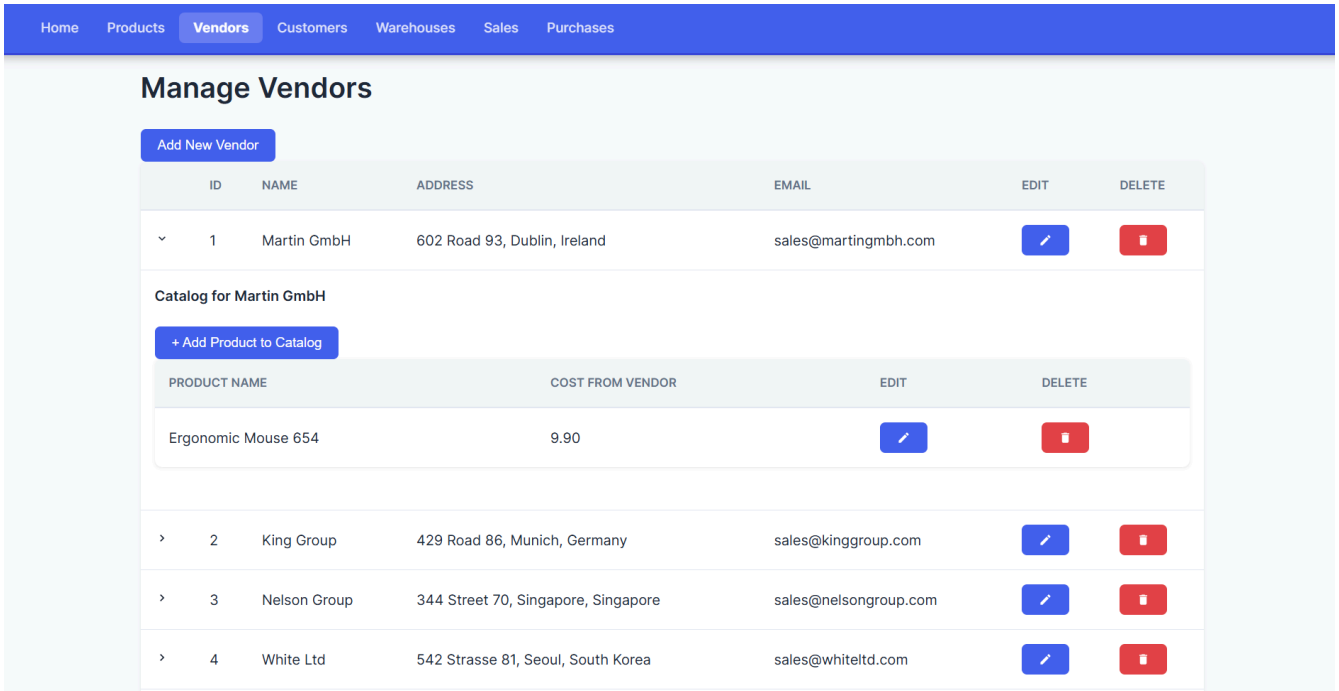
Products (CREATE/READ/UPDATE/DELETE Products):



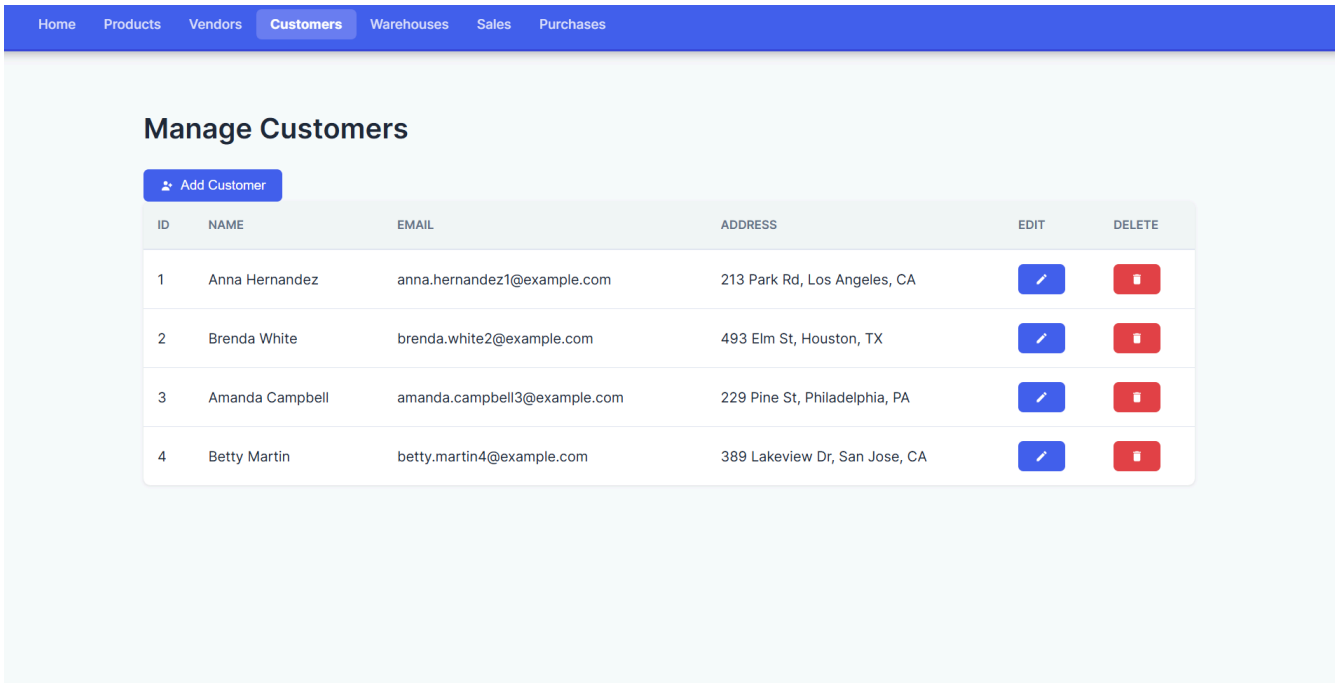
Vendors & VendorProducts (CREATE/READ/UPDATE/DELETE):

Note - VendorProducts (inner table) has M:N CUD operations

Modifies FKs vendorID & productID

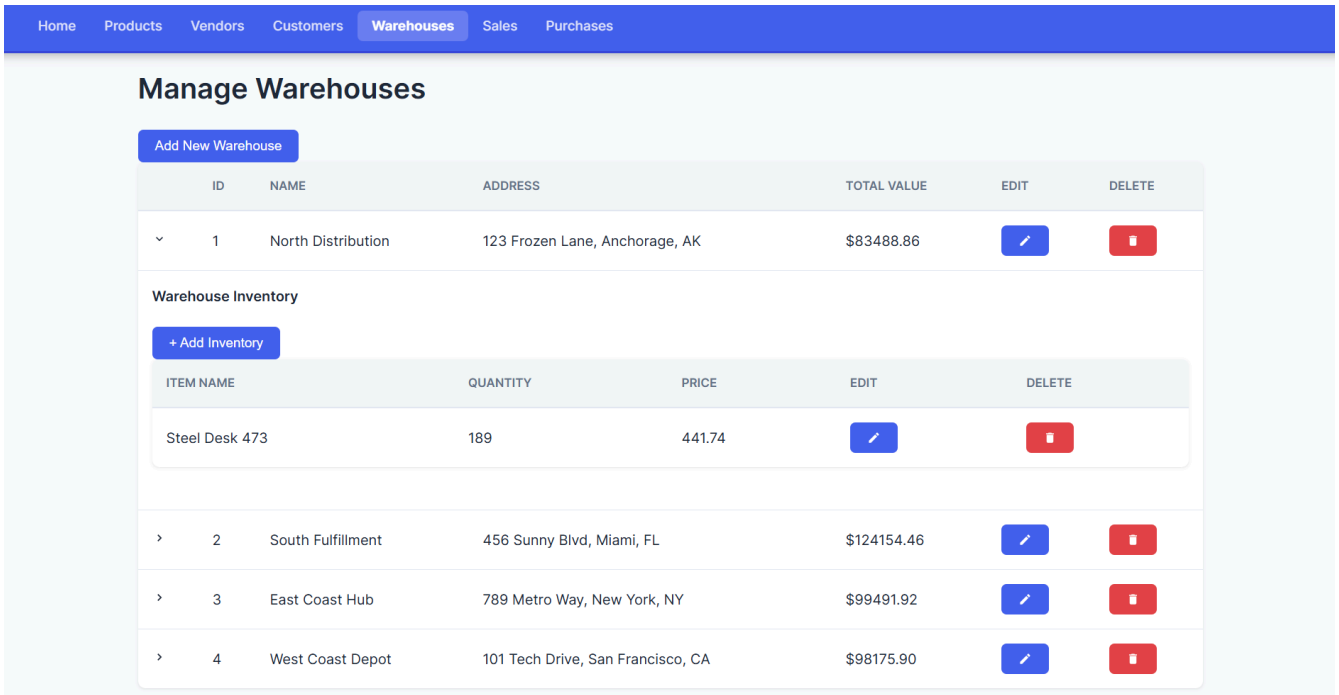


Customers (CREATE/READ/UPDATE/DELETE Customers):



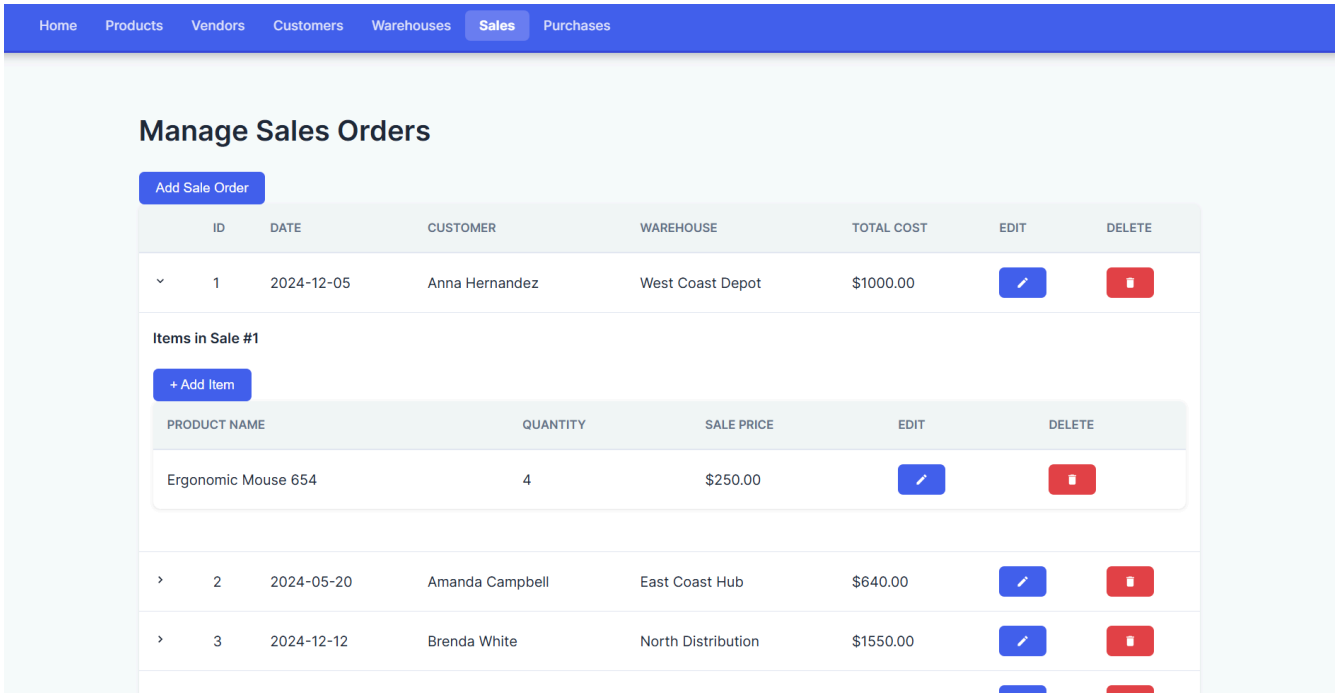
Warehouses & Inventory (CREATE/READ/UPDATE/DELETE):

Note - Inventory (inner table) has M:N CUD operations



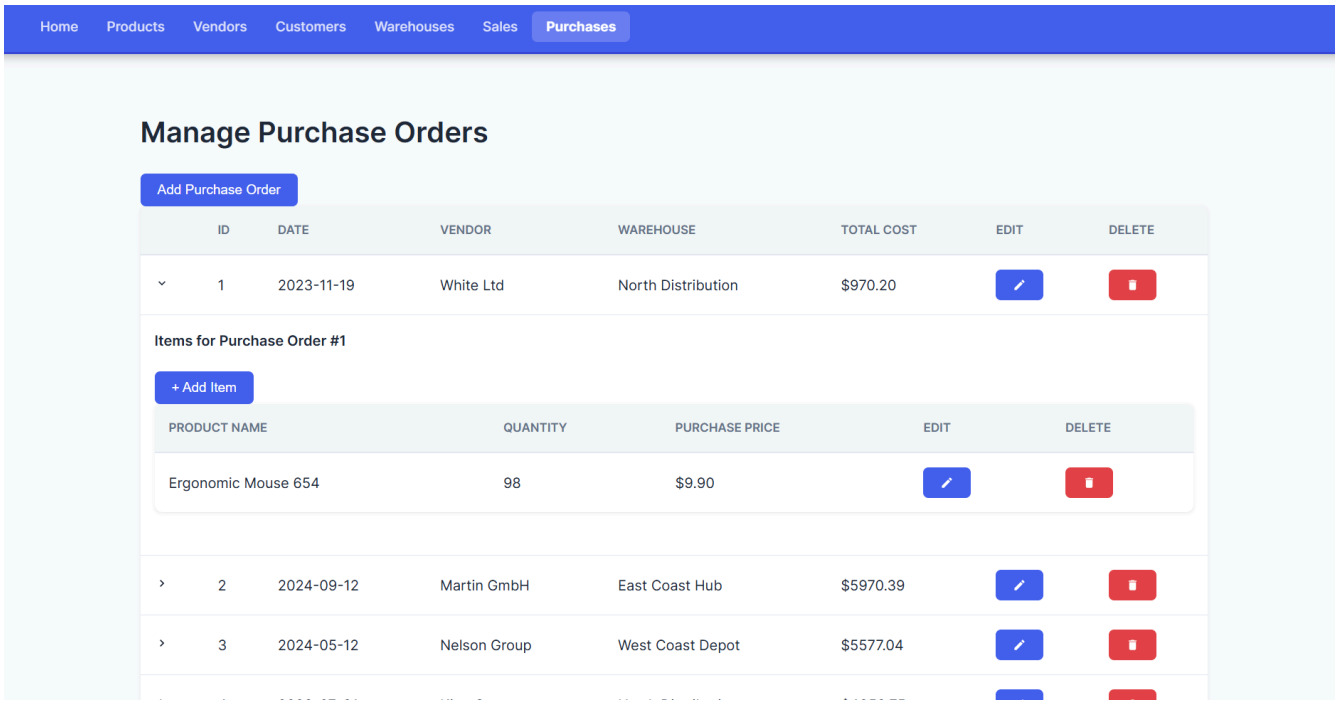
Sales & Sale Order Items (CREATE/READ/UPDATE/DELETE):

Note - Sale Order Items (inner table) has M:N CUD operations



Purchases & Purchase Order Items (CREATE/READ/UPDATE/DELETE):

Note - Purchase Order Items (inner table) has M:N CUD operations



NOTES:

- All create and update operations use forms at the top of the page that open when you click the button
- Expand the row to see specific data in the Warehouses, Vendors, Sales, Purchases pages
- Warehouse inventory is automatically updated on sale/purchase using a trigger